

Course: CSE710 (Advanced Artificial Intelligence), Section-1
Instructor: Dr. Md. Golam Rabiul Alam
Full Marks: 20
Time: 2 hours

Note: There are four questions, answer ALL of them. Course Outcome (CO), Cognitive Level and Mark of each question are mentioned at the right margin.

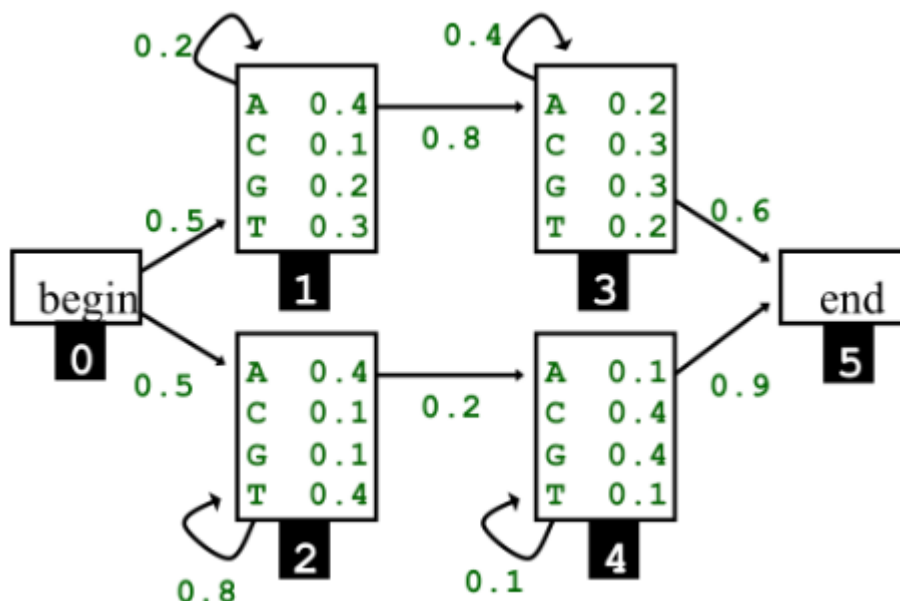
1. Consider a sample dataset $X = \{1, 2, 3, 4, 5\}$ from a population of Normal or Gaussian distribution. We want to estimate the population mean (μ) and standard deviation (σ) from the sample data using Maximum Likelihood Estimation (MLE). The log-likelihood of a Gaussian distribution is as follows: [CO1, C4, Marks: 5]

$$\text{Log}(X | \mu, \sigma) =$$

$$-\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \sum_{j=1}^n (x_j - \mu)^2$$

Where n is the number of elements in X and $j=1$ to n . **Infer** the value of μ and σ within the range 0 to 3 [integer values only] by estimating the Maximum Log-Likelihood of Gaussian distribution. Exceptionally, to skip from the divided by zero error consider 0 as 0.00001.

2. [CO2, C3, Marks: 8]



Use this HMM for answering the Question 2 and Question 3.

Apply the forward algorithm to infer the probability of observing the sequence **AGTT**. Show each step of the recursion and fill the following 2D matrix of forward probabilities $\alpha_{\mathbf{k}(t)}$, where k represents the states and t represents the time stamp.

α	0	1	2	3	4	5
0	1.00	0.00	0.00	0.00	0.00	0.00
A 1	0.00					
G 2	0.00					
T 3	0.00					
T 4	0.00					

3.

Infer the most likely path through the states of the HMM for the sequence **TATA**. Again, show the steps of your algorithm. Your answer should indicate the optimal path as well as the probability of that path.

[CO3, C4,
Marks:7]